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OXIDE SEMICONDUCTOR THIN-FILM TRANSISTOR ARRAY SUBSTRATE AND STRETCHABLE DISPLAY DEVICE INCLUDING SAME

Abstract

An oxide semiconductor thin-film transistor array substrate and a stretchable display device including the same are provided. The stretchable display device can include a substrate having a pixel area and a stretchable area, an insulating member disposed over the substrate in the pixel area and including a first open area exposing the stretchable area, and a thin-film transistor disposed over an area including a side surface of the insulating member exposed by the first open area. Accordingly, the area occupied by the thin-film transistor can be reduced, thereby the size of the pixel area can be reduced and resolution can be improved.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Korean Patent Application No. 10-2024-0025980, filed in the Republic of Korea on Feb. 22, 2024, the entirety of which is hereby expressly incorporated by reference into the present application.

BACKGROUND

Technical Field

[0002] Embodiments of the present disclosure relate to an oxide semiconductor thin-film transistor (TFT) array substrate and a stretchable display device including the same.

Description of the Related Art

[0003] With the development of the information society, demand for image display devices is increasing. As such, a range of display devices, such as liquid crystal display (LCD) devices and organic light-emitting diode display devices, have come into use.

[0004] Display devices can be implemented in a variety of forms, depending on the function to be provided to users. For example, display devices can be implemented in a non-fixed form, such as a foldable display device or a rollable display device. In addition, display devices can be implemented in a stretchable form that can change the shape or size depending on the condition in which display devices are placed or an external force applied to display devices. Since stretchable display devices are stretchable, implementation of display devices having various shapes can further be facilitated.

BRIEF SUMMARY OF THE DISCLOSURE

[0005] Various aspects of the present disclosure provide an oxide semiconductor thin-film transistor (TFT) array substrate able to improve resolution and a stretchable display device including the same.

[0006] Also aspects of the present disclosure provide an oxide semiconductor TFT array substrate able to improve stretching characteristics and a stretchable display device including the same.

[0007] Also aspects of the present disclosure provide an oxide semiconductor TFT array substrate able to reduce device characteristics deterioration due to stretching stress, and a stretchable display device including the same.

[0008] The objectives of the present disclosure are not limited to the aforementioned description, and other objectives not explicitly disclosed herein will be clearly understood by those having ordinary knowledge in the technical field to which the present disclosure pertains from the description provided hereinafter.

[0009] A stretchable display device according to embodiments of the present disclosure includes a substrate including a pixel area and a stretchable area; an insulating member disposed over the substrate in the pixel area and including a first open area exposing the stretchable area; and a thin film transistor (TFT) disposed over an area including a side surface of the insulating member exposed by the first open area.

[0010] A stretchable display device according to embodiments of the present disclosure includes a substrate including a pixel area, a stretchable area, and a pad area; a buffer layer disposed over the

substrate; a first active pattern disposed over the pixel area of the buffer layer, a first gate insulating layer disposed over the buffer layer and covering the first active pattern; a first gate electrode disposed over the first gate insulating layer to overlap a portion of the first active pattern; an interlayer insulating layer disposed over the first gate insulating layer and covering the first gate electrode; a first open area penetrating the interlayer insulating layer, the first gate insulating layer, and the buffer layer to expose the stretchable area of the substrate; a first electrode disposed over the pixel area of the substrate, with at least a portion thereof being exposed by the first open area; a second electrode disposed over the interlayer insulating layer above the first electrode to be adjacent to the first open area; a second active pattern disposed over the second electrode and extending to the first electrode along a side surface of the interlayer insulating layer, a side surface of the first gate insulating layer, and a side surface of the buffer layer exposed by the first open area; a second gate insulating layer disposed over the interlayer insulating layer, covering the second active pattern, and including a second open area exposing the stretchable area of the substrate; and a second gate electrode disposed over the second gate insulating layer to overlap the second active pattern.

[0011] A TFT array substrate according to embodiments of the present disclosure includes a buffer layer disposed over a substrate; a first gate insulating layer disposed over the buffer layer, an interlayer insulating layer disposed over the first gate insulating layer; a first open area penetrating the interlayer insulating layer, the first gate insulating layer, and the buffer layer to expose a portion of the substrate; a first TFT; and a second TFT. The first TFT can include a first oxide semiconductor pattern disposed over the buffer layer and covered with the first gate insulating layer; the first gate insulating layer, and a first gate electrode disposed over the first gate insulating layer to overlap a portion of the first oxide semiconductor pattern. The second TFT can include a first electrode disposed over the substrate, with at least a portion thereof being exposed by the first open area; a second electrode disposed over the interlayer insulating layer above the first electrode to be adjacent to the first open area; a second oxide semiconductor pattern disposed over the second electrode, and extending to the first electrode along a side surface of the interlayer insulating layer, a side surface of the first gate insulating layer, and a side surface of the buffer layer exposed by the first open area; a second gate insulating layer disposed over the interlayer insulating layer and covering the second oxide semiconductor pattern; and a second gate electrode disposed over the second gate insulating layer to overlap the second oxide semiconductor pattern.

[0012] According to embodiments of the present disclosure, in the oxide semiconductor TFT array substrate and the stretchable display device including the same, a TFT can be disposed on a side surface of the insulating member exposed by the first open area exposing the stretchable region to reduce the area occupied by the TFT, thereby reducing the size of the pixel area and improving resolution.

[0013] According to embodiments of the present disclosure, in the oxide semiconductor TFT array substrate and the stretchable display device including the same, the size of the pixel area can be reduced by reducing the area occupied by the TFT, and the free area resulting from the reduced pixel area can be used as a stretchable area to improve stretching characteristics.

[0014] According to embodiments of the present disclosure, in the oxide semiconductor TFT array substrate and the stretchable display device including the same, the impact of stretching stress on the TFT can be reduced to reduce damage to the TFT and reduce or prevent characteristics deterioration of the TFT.

[0015] According to embodiments of the present disclosure, in the oxide semiconductor TFT array substrate and the stretchable display device including the same, the oxide semiconductor can be conductorized by diffusing ions contained in the insulating member to form a conductorization region, thereby reducing the number of process steps. Accordingly, the effect of process optimization can be obtained.

[0016] The effects of the present disclosure are not limited to the aforementioned description, and

other effects not explicitly disclosed herein will be clearly understood by those having ordinary knowledge in the technical field, to which the present disclosure pertains, from the description provided hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] The above and other objectives, features, and advantages of the present disclosure will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:

[0018] FIG. 1 illustrates a schematic system configuration of a stretchable display device according to embodiments of the present disclosure;

[0019] FIG. 2 is a perspective view schematically illustrating an active area of a stretchable display device according to embodiments of the present disclosure;

[0020] FIGS. 3 and 4 are cross-sectional views each illustrating a TFT of an oxide semiconductor TFT array substrate included in a stretchable display device according to embodiments of the present disclosure;

[0021] FIG. 5 is a cross-sectional view illustrating an oxide semiconductor TFT array substrate included in a stretchable display device according to embodiments of the present disclosure;

[0022] FIGS. 6 and 7 are cross-sectional views each illustrating a second TFT of an oxide semiconductor TFT array substrate included in a stretchable display device according to embodiments of the present disclosure; and

[0023] FIGS. 8A to 8F are cross-sectional views illustrating a method of manufacturing an oxide semiconductor TFT array substrate included in a stretchable display device according to embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0024] In the following description of examples or embodiments of the present disclosure, reference will be made to the accompanying drawings in which it is shown by way of illustration specific examples or embodiments that can be implemented, and in which the same reference numerals and signs can be used to designate the same or like components even when they are shown in different accompanying drawings from one another.

[0025] Further, in the following description of examples or embodiments of the present disclosure, detailed descriptions of well-known functions and components incorporated herein will be omitted when it is determined that the description can make the subject matter in some embodiments of the present disclosure rather unclear. The terms such as “including”, “having”, “containing”, “constituting” “made up of”, and “formed of” used herein are generally intended to allow other components to be added unless the terms are used with the term “only”. As used herein, singular forms are intended to include plural forms unless the context clearly indicates otherwise.

[0026] Terms, such as “first”, “second”, “A”, “B”, “(A)”, or “(B)” can be used herein to describe elements of the present disclosure. Each of these terms is not used to define essence, order, sequence, or number of elements etc., but is used merely to distinguish the corresponding element from other elements.

[0027] When it is mentioned that a first element “is connected or coupled to”, “contacts or overlaps” etc. a second element, it should be interpreted that, not only can the first element “be directly connected or coupled to” or “directly contact or overlap” the second element, but a third element can also be “interposed” between the first and second elements, or the first and second elements can “be connected or coupled to”, “contact or overlap”, etc. each other via a fourth element. Here, the second element can be included in at least one of two or more elements that “are connected or coupled to”, “contact or overlap”, etc. each other.

[0028] When time relative terms, such as “after”, “subsequent to”, “next”, “before”, and the like, are used to describe processes or operations of elements or configurations, or flows or steps in operating, processing, manufacturing methods, these terms can be used to describe non-consecutive or non-sequential processes or operations unless the term “directly” or “immediately” is used together.

[0029] In addition, when any dimensions, relative sizes etc. are mentioned, it should be considered that numerical values for an elements or features, or corresponding information (e.g., level, range, etc.) include a tolerance or error range that can be caused by various factors (e.g., process factors, internal or external impact, noise, etc.) even when a relevant description is not specified. Further, the term “can” fully encompasses all the meanings of the term “may”.

[0030] Hereinafter, a variety of embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. All the components of each display device according to all embodiments of the present disclosure are operatively coupled and configured.

[0031] FIG. 1 illustrates a schematic system configuration of a stretchable display device according to embodiments of the present disclosure.

[0032] Referring to FIG. 1, a stretchable display device **100** can include a display panel **110** including an active area AA and a non-active area NA, as well as, a gate driver circuit **120**, a data driver circuit **130**, a controller **140**, and the like for driving the display panel **110**.

[0033] A plurality of gate lines GL and a plurality of data lines DL can be arranged on the display panel **110**, and a plurality of subpixels SP can be disposed in areas where the gate lines GL intersect the data lines DL.

[0034] The gate driver circuit **120** is controlled by the controller **140**, and sequentially outputs scanning signals to the gate lines GL arranged on the display panel **110** to control the driving timing of the subpixels SP.

[0035] The gate driver circuit **120** can include one or more gate driver integrated circuits (GDICs), which can be located on only a first side or on opposite sides of the display panel **110**, depending on the driving system.

[0036] Each of the gate driver integrated circuits can be connected to a bonding pad on the display panel **110** by a tape-automated bonding (TAB) method or a chip-on-glass (COG) method, can be implemented by a gate-in-panel (GIP) method to be disposed directly on the display panel **110**, or in some cases can be integrated on the display panel **110**. In addition, each of the gate driver integrated circuits can be implemented by a chip-on-film (COF) method to be mounted on a film connected to the display panel **110**.

[0037] The data driver circuit **130** receives image data from the controller **140**, and converts the received image data into analog data voltages Vdata. Thereafter, the data driver circuit **130** outputs the data voltages Vdata to the data lines DL through the gate lines GL, respectively, according to the application timing of the scanning signal, so that each of the subpixels SP expresses a brightness corresponding to the image data.

[0038] The data driver circuit **130** can include one or more source driver integrated circuits (SDICs).

[0039] Each of the source driver integrated circuits can include a shift register, a latch circuit, a digital-to-analog converter (DAC), an output buffer, and the like.

[0040] Each of the source driver integrated circuits (SDICs) can be connected to a bonding pad on the display panel **110** by a TAB method or a COG method, can be disposed directly on the display panel **110**, or in some cases can be integrated on the display panel **110**. In addition, each of the source driver integrated circuits can be implemented by a COF method to be mounted on a film connected to the display panel **110**, in which case each of the source driver integrated circuits can be mounted on a circuit film connected to the display panel **110**, and be electrically connected to the display panel **110** through signal lines on the circuit film.

[0041] The controller **140** supplies various control signals to the gate driver circuit **120** and the data

driver circuit **130**, and controls the operation of the gate driver circuit **120** and the data driver circuit **130**.

[0042] The controller **140** can be mounted on a printed circuit board, a flexible printed circuit, or the like, and electrically connected to the gate driver circuit **120** and the data driver circuit **130** through the printed circuit board, the flexible printed circuit, or the like.

[0043] The controller **140** controls the gate driver circuit **120** to output the scanning signal according to the timing realized for each frame, converts externally received image data into a data signal format readable by the data driver circuit **130**, and outputs the converted image data to the data driver circuit **130**.

[0044] The controller **140** receives various timing signals, including a vertical synchronization signal Vsync, a horizontal synchronization signal Hsync, an input data enable signal DE, a clock CLK, and the like, together with the image data from an external source (e.g., a host system).

[0045] The controller **140** generates various control signals using various timing signals received from the external source and delivers the control signals to the gate driver circuit **120** and the data driver circuit **130**.

[0046] In an example, the controller **140** outputs various gate control signals GCS, including a gate start pulse GSP, a gate shift clock GSC, a gate output enable signal GOE, and the like, to control the gate driver circuit **120**.

[0047] The gate start pulse GSP controls the timing at which the gate driver integrated circuits of the gate driver circuit **120** start operating. The gate shift clock GSC is a clock signal input to the gate driver integrated circuits in common to control the shift timing of the scanning signal. The gate output enable signal GOE specifies timing information of the gate driver integrated circuits.

[0048] The controller **140** also outputs various data control signals DCS, including a source start pulse SSP, a source sampling clock SSC, a source output enable signal SOE, and the like, to control the data driver circuit **130**.

[0049] Here, the source start pulse SSP controls the timing at which the source driver integrated circuits of the data driver circuit **130** start data sampling. The source sampling clock SSC is a clock signal that controls the timing of the data sampling by each of the source driver integrated circuits. The source output enable signal SOE controls the output timing of the data driver circuit **130**.

[0050] The stretchable display device **100** can further include a power management circuit that supplies various voltages or currents to, or controls various voltages or currents to be supplied to, the display panel **110**, the gate driver circuit **120**, the data driver circuit **130**, and the like.

[0051] Each of the subpixels SP can be an area defined by the intersection of a gate line GL and a data line DL, and the stretchable display device **100** can be provided with emitting elements ED. In an example, when the stretchable display device **100** is an organic light-emitting display device, an organic light-emitting diode (OLED) and a plurality of circuit elements can be disposed in each of the subpixels SP. In addition, by controlling the current supplied to the OLEDs disposed in the subpixels SP, each of the subpixel SPs can express a brightness corresponding to the image data. In another example, in some cases, the emitting element ED disposed in each of the subpixels SP can be a light-emitting diode (LED) or a micro light-emitting diode (μ LED).

[0052] In addition to the emitting element ED, circuit elements such as thin film transistors can be disposed in the subpixel SP.

[0053] In an example, as shown in FIG. 1, each of the subpixels SP can include two TFTs and one capacitor in addition to the emitting element ED.

[0054] A switching transistor SWT can be electrically connected to the data line DL and a first node N1. The switching transistor SWT can be controlled by a scanning signal supplied to the gate line GL, and can control the supply of a data voltage Vdata to the first node N1, which is the gate node of a driving transistor DRT.

[0055] The driving transistor DRT can be electrically connected to the first driving voltage line DVL1 through which the first driving voltage Vdd is supplied and the emitting element ED. Here,

the first driving voltage Vdd can be a high-potential driving voltage.

[0056] The driving transistor DRT can be controlled by the data voltage Vdata supplied to the first node N1, and can control the current supplied to the emitting element ED.

[0057] Both the switching transistor SWT and the driving transistor DRT are shown implemented as P-type transistors for example, but can be implemented as N-type transistors in some cases.

[0058] The storage capacitor Cstg can be electrically connected to the first node N1 and a second node N2, and can maintain the data voltage Vdata for a single frame.

[0059] The emitting element ED can be electrically connected to the driving transistor DRT and the second drive voltage line DVL2 through which a second drive voltage Vss is supplied. Here, the second driving voltage Vss can be a low-potential driving voltage.

[0060] The emitting element ED can emit light according to the current supplied through the driving transistor DRT. That is, the emitting element ED can express a brightness according to the current supplied through the driving transistor DRT according to the data voltage Vdata supplied to the first node N1, and can express a luminance according to the image data. In addition, the subpixel SP can further include a TFT or a capacitor depending on the type.

[0061] The stretchable display device **100** described above can display an image in a state in which stretchable display device **100** is stretched by an external force.

[0062] FIG. 2 is a perspective view schematically illustrating an active area of a stretchable display device according to embodiments of the present disclosure.

[0063] Referring to FIG. 2, the stretchable display device **100** can include a first substrate **10A** and a second substrate **10B**.

[0064] The first substrate **10A** is a substrate for supporting and protecting the various components of the stretchable display device **100**. The second substrate **10B** is a substrate for covering and protecting the various components of the stretchable display device **100**.

[0065] The first substrate **10A** and the second substrate **10B** can be formed of an insulating material that is bendable or stretchable. For example, the first substrate **10A** and the second substrate **10B** can include polyimide (PI), polyacrylate, polyacetate, silicone rubber such as polydimethylsiloxane (PDMS), or elastomer such as polyurethane (PU) or polytetrafluoroethylene (PTFE), and thus can have flexible properties. The material of the first substrate **10A** and the material of the second substrate **10B** can be the same, but can be varied without being limited thereto.

[0066] A plurality of pixel areas PIXA in each of which an emitting element ED and the like can be arranged in an active area of the first substrate **10A** can be disposed to be spaced apart from each other. Pad areas can be provided in a non-active area of the first substrate **10A**.

[0067] The pixel areas PIXA and the pad areas are areas having a fixed shape and can be regarded as rigid areas. The pixel areas PIXA and the pad areas can maintain a constant shape even in the case that the stretchable display device **100** is in a stretched state.

[0068] Areas other than the pixel areas PIXA and the pad areas are stretchable and can be considered as stretchable areas. A plurality of signal lines SL can be disposed in the stretchable areas.

[0069] The signal lines SL can be electrically connected to elements disposed in the pixel areas PIXA and/or pads disposed in the pad areas. The signal lines SL can be disposed, for example, in a bent form in the stretchable areas between the pixel areas PIXA. Thus, the length of a signal lines SL disposed between the pixel areas PIXA can be greater than a straight distance between adjacent pixel areas PIXA. In another example, the signal lines SL can be disposed in a bent form in the stretchable areas between the pixel areas PIXA and the pad areas. Thus, the length of the signal lines SL disposed between the pixel areas PIXA and the pad areas can be greater than the straight distance (e.g., direct distance) between a pixel area PIXA and an adjacent pad area.

[0070] When the stretchable display device **100** is in a stretched state, the distance between the pixel areas PIXA and/or the distance between the pixel areas PIXA and the pad areas can increase. That is, when an external force is applied to the stretchable display device **100**, the pixel areas

PIXA and the pad areas can maintain a fixed shape, and only the shape of the stretchable areas can change. This can also be regarded as the area change rate of the stretchable area is greater than the area change rate of the pixel areas PIXA and pad areas.

[0071] As the signal lines SL are disposed in a bent form, when the stretchable display device **100** is in a stretched state, the signal lines SL can be stretched to perform the signal supplying function. Accordingly, the stretchable display device **100** can perform display driving while maintaining constant performance in both the unstretched base state and the stretched state.

[0072] FIG. **3** is a cross-sectional view illustrating a TFT of an oxide semiconductor TFT array substrate included in a stretchable display device according to embodiments of the present disclosure.

[0073] Referring to FIG. **3**, an insulating member **20** having a first open area OP**11** exposing a stretchable area FA can be disposed over the first substrate **10A** in the pixel area PIXA. A multi-buffer layer **11** can be disposed between the insulating member **20** and the first substrate **10A**.

[0074] The multi-buffer layer **11** can have a structure, for example, in which a plurality of thin films are stacked together. For example, the multi-buffer layer **11** can have a structure in which silicon nitride (SiNx) and silicon oxide (SiOx) are alternately stacked. In another example, the multi-buffer layer **11** can have a structure in which organic and inorganic films are alternately stacked. The multi-buffer layer **11** can serve to retard upward diffusion of moisture and/or oxygen. In another example, the multi-buffer layer **11** can be omitted.

[0075] The insulating member **20** can include an inorganic material, and can have a single-layer structure or a multiple-layer structure. The insulating member **20** can be formed of, for example, silicon oxide (SiOx).

[0076] A thin-film transistor (TFT) **30** can be disposed on an area including a side surface of the insulating member **20** exposed by the first open area OP**11**. The TFT **30** can be one of the transistors included in a subpixel. For example, the TFT **30** can be one of the switching transistors and the driving transistors described with reference to FIG. **1**.

[0077] The TFT **30** can include a first electrode **31**, a second electrode **32**, an active pattern **33**, a gate insulating layer **34**, and a gate electrode **35**.

[0078] The first electrode **31** can be disposed over the multi-buffer layer **11** in the pixel area PIXA. At least a portion of the first electrode **31** can be exposed by the first open area OP**11** of the insulating member **20**. The remaining portions of the first electrode **31** can be covered with the insulating member **20**.

[0079] The second electrode **32** can be disposed over the insulating member **20** above the first electrode **31** adjacent to the first open area OP**11**. One of the first electrode **31** and the second electrode **32** can be the source electrode of the TFT **30**, and the other can be the drain electrode of the TFT **30**.

[0080] The active pattern **33** can be disposed on the second electrode **32** and extend to the first electrode **31** along the side surface of the insulating member **20** exposed by the first open area OP**11**. The top portion of the active pattern **33** can be in contact with the second electrode **32** on the second electrode **32**, and the bottom portion of the active pattern **33** can be over the first electrode **31** and in contact with the first electrode **31**.

[0081] The active pattern **33** can be formed of an oxide semiconductor. The active pattern **33** can be an oxide semiconductor pattern.

[0082] The oxide semiconductor can include an oxide of a metal, such as zinc (Zn), indium (In), gallium (Ga), tin (Sn), or titanium (Ti), or a combination of such a metal and an oxide thereof. More specifically, oxide semiconductors can include zinc oxide (ZnO), zinc-tin oxide (ZTO), zinc-indium oxide (ZIO), indium oxide (InO), titanium oxide (TiO) indium-gallium-zinc oxide (IGZO), indium-zinc-tin oxide (IZTO), and the like. The active pattern **33** can have semiconductor properties.

[0083] The gate insulating layer **34** can be disposed on the top surface of the insulating member **20**

and on the side surface of the insulating member **20** exposed by the first open area **OP11**, and can cover the active pattern **33**. The gate insulating layer **34** can have a second open area **OP12** exposing the stretchable area **FA** of the first substrate **10A**.

[0084] The gate electrode **35** can be disposed over the gate insulating layer **34** to overlap the active pattern **33**. A portion of the gate electrode **35** can overlap a portion of the active pattern **33** disposed over the side surface of the insulating member **20** and the top surface of the first electrode **31** exposed by the first open area **OP11**, and another portion of the gate electrode **35** can overlap another portion of the active pattern **33** disposed over the second electrode **32**.

[0085] A passivation layer **60** can be disposed over the gate insulating layer **34** to cover the gate electrodes **35**. The passivation layer **60** can have a third open area **OP13** that exposes the stretchable area **FA** of the first substrate **10A**. A side surface of the passivation layer **60** can be exposed by the third open area **OP13**.

[0086] The passivation layer **60** can be implemented as a single layer or a plurality of layers, and can be formed of an organic material. For example, the passivation layer **60** can include, but is not limited to, an acryl-based organic material.

[0087] A signal line **70** can be disposed over the passivation layer **60**. The signal line **70** can extend to the stretchable area **FA** of the first substrate **10A** along the side surface of the passivation layer **60** exposed by the third open area **OP13**. The signal line **70** can be electrically connected to at least one of the components disposed in the pixel area **PIXA**. The signal line **70** represents a component corresponding to the signal line **SL** described with reference to FIG. 2.

[0088] In this manner, by disposing the TFT **30** on the area including the side surface of the insulating member **20** exposed by the first open area **OP11** that opens the stretchable area **FA** of the first substrate **10A**, the area occupied by the TFT **30** can be reduced without reducing the effective channel length of the TFT **30**. Accordingly, the resolution can be improved by reducing the size of the pixel area.

[0089] In addition, the free area resulting from the reduction in the size of the pixel area **PIXA** can be used as a stretchable area **FA** to increase the size of the stretchable area **FA**, thereby improving the stretching characteristics.

[0090] Furthermore, since the active pattern **33**, the gate insulating layer **34**, and the gate electrode **35** of the TFT **30** are arranged in a direction different from the direction in which the stretchable display device is stretched, the impact of the stretching stress on the TFT can be reduced, thereby reducing damage to the TFT **30** and reducing or preventing characteristic deterioration of the TFT.

[0091] FIG. 4 is a cross-sectional view illustrating a TFT of an oxide semiconductor TFT array substrate included in a stretchable display device according to embodiments of the present disclosure.

[0092] Referring to FIG. 4, the active pattern **33** of the TFT **30** can be formed of an oxide semiconductor. The active pattern **33** can be an oxide semiconductor pattern. The active pattern **33** can include a first conductorization region **33A**, a second conductorization region **33C**, and a first oxide semiconductor region **33D**.

[0093] The first conductorization region **33A** can be disposed over the first electrode **31** and in direct contact with the first electrode **31**.

[0094] The second conductorization region **33C** can be disposed adjacent to the second electrode **32**. The first oxide semiconductor region **33D** can be disposed between the first conductorization region **33A** and the second conductorization region **33C**.

[0095] The first oxide semiconductor region **33D** can have semiconductor properties. The first oxide semiconductor region **33D** can be formed of an oxide semiconductor. The first conductorization region **33A** and the second conductorization region **33C** can be regions in which the oxide semiconductor is conductorized (i.e., becomes conductive) by hydrogen diffused from the insulating member **20**.

[0096] The insulating member **20** can include a first insulating layer **20A** disposed over the multi-

buffer layer **11** and the first electrode **31**, a second insulating layer **20C** disposed over the first insulating layer **20A**, and a third insulating layer **20B** disposed over the second insulating layer **20C**.

[0097] Each of the first insulating layer **20A**, the second insulating layer **20C**, and the third insulating layer **20B** can be formed of an insulating material, but the first insulating layer **20A** and the third insulating layer **20B** can be formed of a material having a different hydrogen content than the second insulating layer **20C**. The first insulating layer **20A** and the third insulating layer **20B** can be formed of a high hydrogen film material having a hydrogen content equal to or higher than a threshold value, and the second insulating layer **20C** can be formed of a low hydrogen film material having a hydrogen content lower than the threshold value. Each of the first insulating layer **20A** and the third insulating layer **20B** can have a higher hydrogen content than the second insulating layer **20C**.

[0098] In a subsequent process, the hydrogen contained in the first insulating layer **20A**, the second insulating layer **20C**, and the third insulating layer **20B** can diffuse into the oxide semiconductor in the active pattern **33**. A large amount of hydrogen from the first insulating layer **20A** and the third insulating layer **20B** having a high hydrogen content can diffuse into the oxide semiconductor, thereby conductorizing the oxide semiconductor. Even in the case that hydrogen diffuses from the second insulating layer **20C** having a low hydrogen content into the oxide semiconductor, the oxide semiconductor can retain semiconductor properties instead of being conductorized. The threshold value refers to a minimum value or a value selected from a certain range of hydrogen content that enables the oxide semiconductor of the active pattern **33** to be conductorized.

[0099] For example, each of the first insulating layer **20A** and the third insulating layer **20B** can be formed of at least one selected from the group of materials including silicon nitride (SiN_x), aluminum oxide (AlO_x), hafnium oxide (HfO_x), zirconium oxide (ZrO_x), tantalum oxide (TaO_x), or the like. The second insulating layer **20C** can be formed of silicon oxide (SiO_x).

[0100] Here, the silicon nitride (SiN_x) can be formed using silane (SiH_4) and ammonia (NH_3) as source gases. Aluminum oxide (AlO_x) can be formed using trimethyl aluminum (TMA, $\text{C}_3\text{H}_9\text{Al}$) as a source gas. Hafnium oxide (HfO_x) can be formed using CpHf ($\text{C}_{11}\text{H}_{23}\text{N}_3\text{Hf}$) as a source gas. Zirconium oxide (ZrO_x) can be formed using CpZr ($\text{C}_{11}\text{H}_{23}\text{N}_3\text{Zr}$) as a source gas. Tantalum oxide (TaO_x) can be formed using CpTa ($\text{C}_{17}\text{H}_{39}\text{N}_4\text{Ta}$) as a source gas.

[0101] In response to the diffusion of hydrogen from the first insulating layer **20A**, which is a high hydrogen film, the oxide semiconductor in a portion of the active pattern **33** in contact with the side surface of the first insulating layer **20A** can be conductorized, thereby forming the first conductorization region **33A**. The first conductorization region **33A** is a region that is conductorized by the diffusion of hydrogen contained in the first insulating layer **20A** into the oxide semiconductor, and can contact the side surface of the first insulating layer **20A**.

[0102] In response to the diffusion of hydrogen from the third insulating layer **20B**, which is a high hydrogen film, the oxide semiconductor in another portion of the active pattern **33** in contact with a side surface of the third insulating layer **20B** can be conductorized, thereby forming the second conductorization region **33C**. The second conductorization region **33C** is a region that is conductorized by the diffusion of hydrogen contained in the third insulating layer **20B** into the oxide semiconductor, and can contact the side surface of the third insulating layer **20B**.

[0103] In another portion of the active pattern **33** in contact with the side surface of the second insulating layer **20C**, which is a low hydrogen film, the oxide semiconductor is not conductorized. Accordingly, the first oxide semiconductor region **33D** can be in contact with the side surface of the second insulating layer **20C**.

[0104] As described above, the first conductorization region **33A** and the second conductorization region **33C** having a low resistance can be formed in the active pattern **33** of the TFT **30**, thereby reducing the offset resistance and increasing the electrical conductivity. In addition, in the process

used to fabricate the stretchable display device, the oxide semiconductor of the active pattern **33** can be conductorized by the diffusion of the hydrogen contained in the insulating member **20**, thereby forming the TFT **30** having excellent electrical conductivity without adding any additional process steps.

[0105] The active pattern **33** can further include an extension region **33B**. The extension region **33B** can be disposed on the second electrode **32** and can be in direct contact with the second electrode **32**. The extension region **33B** can have different characteristics depending on the type of metallic material of the second electrode **32**. The extension region **33B** can be an oxide semiconductor region or a conductorization region in which an oxide semiconductor is conductorized.

[0106] In an example, the second electrode **32** can be formed of a metal having a higher oxidizing power than the oxide semiconductor. For example, the second electrode **32** can be formed of at least one of Ti, Al, MoTi, ITO, IZO, or Zr. In such a case, the metal of the second electrode **32** can be combined with the oxygen contained in the oxide semiconductor at the contact portion between the second electrode **32** and the oxide semiconductor, so that the oxide semiconductor can lose oxygen and be conductorized. In such a case, the extension region **33B** can be a conductorization region.

[0107] In another example, the second electrode **32** can be formed of a metal having a lower oxidizing power than the oxide semiconductor. For example, the second electrode **32** can be formed of at least one of Mo, Cu, W, or Fe. In such a case, the metal of the second electrode **32** may not be combined with the oxygen contained in the oxide semiconductor at the contact portion between the second electrode **32** and the oxide semiconductor, so that the oxide semiconductor can remain in the oxide semiconductor state without being conductorized. In such a case, the extension region **33B** can be an oxide semiconductor region.

[0108] FIG. **5** is a cross-sectional view illustrating an oxide semiconductor TFT array substrate included in a stretchable display device according to embodiments of the present disclosure.

[0109] Referring to FIG. **5**, a multi-buffer layer **11** covering the pixel area PIXA and the pad area PADA and exposing the stretchable area FA can be disposed over the first substrate **10A**.

[0110] The multi-buffer layer **11** can have a structure in which, for example, a plurality of thin films are stacked together. For example, the multi-buffer layer **11** can have a structure in which silicon nitride (SiNx) and silicon oxide (SiOx) are alternately stacked. In another example, the multi-buffer layer **11** can have a structure in which organic films and inorganic films are alternately stacked. The multi-buffer layer **11** can serve to retard upward diffusion of moisture and/or oxygen. In another example, the multi-buffer layer **11** can be omitted.

[0111] The first electrode **31** of the second TFT **30** can be disposed over the multi-buffer layer **11** in the pixel area PIXA. The first electrode **31** can be one of the source electrode and the drain electrode of the second TFT **30**.

[0112] In addition, or optionally, a bottom shield metal pattern **91** can be disposed over the multi-buffer layer **11** in the pixel area PIXA. The bottom shield metal pattern **91** can be disposed to overlap a first active pattern **81** of the first TFT **80**. The bottom shield metal pattern **91** can serve to protect the first active pattern **81** of the first TFT **80**. The bottom shield metal pattern **91** can prevent the generation of potential on the surface of the first substrate **10A** and light from entering from the outside.

[0113] The first electrode **31** can be formed of the same material as the bottom shield metal pattern **91**. The first electrode **31** and the bottom shield metal pattern **91** can include, but are not limited to, molybdenum (Mo).

[0114] The insulating member **20** can be disposed over the multi-buffer layer **11**. The insulation member **20** can include an active buffer layer **21**, a first gate insulating layer **22**, and an interlayer insulating layer **23**.

[0115] The active buffer layer **21** can be disposed over the multi-buffer layer **11** to cover the first

electrode **31** and the bottom shield metal pattern **91**. In an embodiment, the active buffer layer **21** can be formed of a low hydrogen film having a hydrogen content lower than a threshold value. For example, the active buffer layer **21** can be formed of silicon oxide (SiOx).

[0116] A first active pattern **81** of the first TFT **80** can be disposed over the active buffer layer **21** in the pixel area PIXA. The first active pattern **81** can include a channel region, and a source region and a drain region on opposite sides of the channel region. The first TFT **80** can be an oxide TFT. In such a case, the first active pattern **81** can be formed of an oxide semiconductor. In another example, the first active pattern **81** can be formed of an amorphous silicon (a-Si), polycrystalline silicon (poly-Si), organic semiconductor, or the like.

[0117] A first gate insulating layer **22** can be disposed over the active buffer layer **21** to cover the first active pattern **81**. In an embodiment, the first gate insulating layer **22** can include a low hydrogen film having a hydrogen content lower than the threshold value. For example, the first gate insulating layer **22** can be formed of silicon oxide (SiOx).

[0118] A first gate electrode **82** of the first TFT **80** can be disposed over the first gate insulating layer **22**. The first gate electrode **82** can overlap the channel region of the first active pattern **81**. In addition or optionally, a capacitor bottom electrode **41** can be disposed over the first gate insulating layer **22** in the pixel area PIXA. In addition or optionally, a first pad layer **51** can be disposed over the first gate insulating layer **22** in the pad area PADA.

[0119] The first gate electrode **82**, the capacitor bottom electrode **41** and the first pad layer **51** can be formed of the same material. The first gate electrode **82**, the capacitor bottom electrode **41**, and the first pad layer **51** can be formed of one of various metallic materials, such as molybdenum (Mo), aluminum (Al), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd), and copper (Cu), an alloy of one or more thereof, or multiple layers thereof, but are not limited thereto.

[0120] The interlayer insulating layer **23** can be disposed over the first gate insulating layer **22** to cover the first gate electrode **82**, the capacitor bottom electrode **41**, and the first pad layer **51**. In an embodiment, the interlayer insulating layer **23** can be formed of a low hydrogen film having a hydrogen content lower than the threshold value. For example, the interlayer insulating layer **23** can be formed of silicon oxide (SiOx).

[0121] A first open area OP11 that penetrates the interlayer insulating layer **23**, the first gate insulating layer **22**, and the active buffer layer **21** to expose the stretchable area FA of the first substrate **10A** can be provided. The first open area OP11 can expose a side surface of the interlayer insulating layer **23**, a side surface of the first gate insulating layer **22**, a side surface of the active buffer layer **21**, and a portion of the first electrode **31**.

[0122] A second electrode **32** of the second TFT **30** can be disposed over the interlayer insulating layer **23** above the first electrode **31**. The second electrode **32** can be disposed adjacent to the first open area OP11. The second electrode **32** can be one of a source electrode or a drain electrode of the second TFT **30**.

[0123] The capacitor top electrode **42** can be disposed over the interlayer insulating layer **23** in the pixel area PIXA. The capacitor top electrode **42** can be disposed to overlap the capacitor bottom electrode **41**.

[0124] A fourth open area OP21 that penetrates the interlayer insulating layer **23** to expose the first pad layer **51** can be provided. The top surface of the first pad layer **51** and the side surface of the interlayer insulating layer **23** can be exposed by the fourth open area OP21. A second pad layer **52** can be disposed over the first pad layer **51** exposed by the fourth open area OP21. The second pad layer **52** can cover the top surface of the first pad layer **51** and the side surface of the interlayer insulating layer **23** exposed by the second open area OP12, and can be in direct contact with the first pad layer **51**.

[0125] The second electrode **32** may, for example, be formed of a metal having a higher oxidizing power than the oxide semiconductor. For example, the second electrode **32** can be formed of at least one of Ti, Al, MoTi, ITO, IZO, or Zr. In another example, the second electrode **32** can be

formed of a metal having a lower oxidizing power than the oxide semiconductor. For example, the second electrode **32** can be formed of at least one of Mo, Cu, W, or Fe. The capacitor top electrode **42** and the second pad layer **52** can be made of the same material as the second electrode **32**.

[0126] The second active pattern **33** of the second TFT **30** can be disposed over the second electrode **32** and can extend to the first electrode **31** along the side surface of the interlayer insulating layer **23**, the side surface of the first gate insulating layer **22**, and the side surface of the active buffer layer **21** exposed by the first open area OP11. A top portion of the second active pattern **33** can be in contact with the top surface and the side surface of the second electrode **32**, and a bottom portion of the second active pattern **33** can be in contact with the top surface of the first electrode **31**.

[0127] A second gate insulating layer **34** can be disposed over the interlayer insulating layer **23** to cover the side surface of the interlayer insulating layer **23**, the side surface of the first gate insulating layer **22**, the side surface of the active buffer layer **21**, and the second active pattern **33** exposed by the first open area OP11.

[0128] A source electrode **83**, a drain electrode **84**, a line (or a conductive line) **85**, the second gate electrode **35**, and the third pad layer **53** can be disposed over the second gate insulating layer **34**.

[0129] The source electrode **83** and the drain electrode **84** can extend through the second gate insulating layer **34**, the interlayer insulating layer **23**, and the first gate insulating layer **22** to be connected to the source region and the drain region of the first active pattern **81**, respectively.

[0130] The line **85** can be electrically connected to at least one of the components disposed in the pixel area PIXA.

[0131] The second gate electrode **35** can be disposed to overlap the second active pattern **33**. A portion of the second gate electrode **35** can overlap the side surface of the insulating member **20** exposed by the first open area OP11 and a portion of the second active pattern **33** disposed over the first electrode **31**, and another portion of the second gate electrode **35** can overlap another portion of the second active pattern **33** disposed over the second electrode **32**.

[0132] The second gate insulating layer **34** can have a second open area OP12 exposing the stretchable area FA of the first substrate **10A**. The second gate insulating layer **34** can have a fifth open area OP22 exposing the second pad layer **52**. A portion of the top surface of the second pad layer **52** and a side surface of the second gate insulating layer **34** can be exposed by the fifth open area OP22.

[0133] A third pad layer **53** can be disposed over the second pad layer **52** exposed by the fifth open area OP22. The third pad layer **53** can cover the top surface of the second pad layer **52** and the side surface of the second gate insulating layer **34** exposed by the fifth open area OP22, and can be in direct contact with the second pad layer **52**.

[0134] The second gate electrode **35** can be formed of the same material as the source electrode **83**, the drain electrode **84**, the line **85**, and the third pad layer **53**. The second gate electrode **35**, the source electrode **83**, the drain electrode **84**, the line **85**, and the third pad layer **53** can be formed of one of various metallic materials, such as molybdenum (Mo), aluminum (Al), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd), and copper (Cu), an alloy of one or more thereof, or multiple layers thereof, but are not limited thereto. A portion of the second gate electrode **35** can be disposed on the same layer as the source electrode **83** and the drain electrode **84**.

[0135] The first active pattern **81**, the first gate insulating layer **22**, the first gate electrode **82**, the source electrode **83**, and the drain electrode **84** can form the first TFT **80**. The first TFT **80** can have a coplanar structure.

[0136] The first electrode **31**, the second electrode **32**, the second active pattern **33**, the second gate insulating layer **34**, and the second gate electrode **35** can form the second TFT **30**. The second TFT **30** can have a vertical structure. Each of the first TFT **80** and the second TFT **30** can be one of the driving transistor and the scanning transistor described with reference to FIG. 1.

[0137] The capacitor bottom electrode **41**, the interlayer insulating layer **23**, and the capacitor top electrode **42** can form a capacitor **30**. The capacitor **30** can be a storage capacitor as described with reference to FIG. **1**.

[0138] The first pad layer **51**, the second pad layer **52**, and the third pad layer **53** can form the pad **50**. The pad **50** can be one of, but is not limited to, a gate pad for delivering a gate voltage to the subpixel, a data pad for delivering a data voltage to the subpixel, or a voltage pad for delivering a low potential voltage or a high potential voltage to the subpixel.

[0139] The passivation layer **60** can be disposed over the second gate insulating layer **34** to cover the source electrode **83** and the drain electrode **84**, the line **85**, the second gate electrode **35**, and the third pad layer **53**.

[0140] The passivation layer **60** can prevent moisture and oxygen from entering the first and second TFT **30**. The passivation layer **60** can be formed of an inorganic material, and can be a single layer or multiple layers, but is not limited thereto.

[0141] The passivation layer **60** can have a third open area **OP13** that exposes the stretchable area **FA** of the first substrate **10A**. Around the stretchable area **FA**, a side surface of the passivation layer **60** can be exposed by the third open area **OP13**.

[0142] The passivation layer **60** can have a sixth open area **OP23** exposing the third pad layer **53**. Around the third pad layer **53**, the side surface of the passivation layer **60** can be exposed by the sixth open area **OP23**.

[0143] A signal line **70** can be disposed on the passivation layer **60**. The signal line **70** can be connected to line **85** through a contact hole that penetrates the passivation layer **60**.

[0144] The signal line **70** can extend to the stretchable area **FA** of the first substrate **10A** along the side surface of the passivation layer **60** exposed by the third open area **OP13**. In addition, the signal line **70** can extend to the pad area **PADA** and can be connected to the pad **50** through the sixth open area **OP23** in the pad area **PAD**.

[0145] Embodiments of the present disclosure illustrate, but are not limited to, a case in which the stretchable area **FA** is disposed between the pixel area **PIXA** and the pad area **PADA**, and the signal line **70** connects the pixel area **PIXA** and the pad area **PADA**. The stretchable area **FA** can be disposed between neighboring pixel areas **PIXA**, and the signal line **70** can connect the neighboring pixel areas **PIXA**.

[0146] FIGS. **6** and **7** are cross-sectional views each illustrating a second TFT of an oxide semiconductor TFT array substrate included in a stretchable display device according to embodiments of the present disclosure.

[0147] Referring to FIG. **6**, the second active pattern **33** of the second TFT **30** can include a first conductorization region **33A**, a second conductorization region **33C**, and a first oxide semiconductor region **33D**.

[0148] The first conductorization region **33A** can be disposed over the first electrode **31** and can be in direct contact with the first electrode **31**.

[0149] The second conductorization region **33C** can be disposed adjacent to the second electrode **32**. The first oxide semiconductor region **33D** can be disposed between the first conductorization region **33A** and the second conductorization region **33C**.

[0150] The first oxide semiconductor region **33D** can have semiconductor properties. The first conductorization region **33A** and the second conductorization region **33C** can be regions in which the oxide semiconductor is conductorized by hydrogen diffused from the insulating member **20**.

[0151] The insulating member **20** can include an active buffer layer **21**, a first gate insulating layer **22**, and an interlayer insulating layer **23**.

[0152] The active buffer layer **21** can include a first level active buffer layer **21A** disposed over the multi-buffer layer **11** and covering the first electrode **31** and a second level active buffer layer **21B** disposed between the first level active buffer layer **21A** and the first gate insulating layer **22**.

[0153] The interlayer insulating layer **23** can include a first level interlayer insulating layer **23A**

disposed over the first gate insulating layer **22** and a second level interlayer insulating layer **23B** disposed between the first level interlayer insulating layer **23A** and the second gate insulating layer **34**.

[0154] Each of the first level active buffer layer **21A**, the second level active buffer layer **21B**, the first gate insulating layer **22**, the first level interlayer insulating layer **23A**, and the second level interlayer insulating layer **23B** can be formed of an insulating material, but the first level active buffer layer **21A** and the second level interlayer insulating layer **23B** can be formed of a material having a different hydrogen content than the second level active buffer layer **21B**, the first gate insulating layer **22**, and the first level interlayer insulating layer **23A**. The first level active buffer layer **21A** and the second level interlayer insulating layer **23B** can be formed of a high hydrogen film material having a hydrogen content equal to or higher than a threshold value, and the second level active buffer layer **21B**, the first gate insulating layer **22**, and the first level interlayer insulating layer **23A** can be formed of a low hydrogen film material having a hydrogen content lower than the threshold value. Each of the first level active buffer layer **21A** and the second level interlayer insulating layer **23B** can have a higher hydrogen content than the second level active buffer layer **21B**, the first gate insulating layer **22**, and the first level interlayer insulating layer **23A**.

[0155] For example, the first level active buffer layer **21A** and the second level interlayer insulating layer **23B** can be formed of at least one selected from the group of materials including silicon nitride (SiNx), aluminum oxide (AlOx), hafnium oxide (HfOx), zirconium oxide (ZrOx), tantalum oxide (TaOx), or the like. The second level active buffer layer **21B**, the first gate insulating layer **22**, and the first level interlayer insulating layer **23A** can be formed of silicon oxide (SiOx).

[0156] Here, the silicon nitride (SiNx) can be formed using silane (SiH.sub.4) and ammonia (NH.sub.3) as source gases. Aluminum oxide (AlOx) can be formed using trimethyl aluminum (TMA, C.sub.3H.sub.9Al) as a source gas. Hafnium oxide (HfOx) can be formed using CpHf (C.sub.11H.sub.23N.sub.3Hf) as a source gas. Zirconium oxide (ZrOx) can be formed using CpZr (C.sub.11H.sub.23N.sub.3Zr) as a source gas. Tantalum oxide (TaOx) can be formed using CpTa (C.sub.17H.sub.39N.sub.4Ta) as a source gas.

[0157] In response to the diffusion of hydrogen from the first level active buffer layer **21A**, which is a high hydrogen film, the oxide semiconductor in a portion of the second active pattern **33** in contact with the side surface of the first level active buffer layer **21A** can be conductorized, thereby forming the first conductorization region **33A**. The first conductorization region **33A** is a region that is conductorized by the diffusion of hydrogen contained in the first level active buffer layer **21A** into the oxide semiconductor, and can contact the side surface of the first level active buffer layer **21A**.

[0158] In response to the diffusion of hydrogen from the second level interlayer insulating layer **23B**, which is a high hydrogen film, the oxide semiconductor in another portion of the second active pattern **33** in contact with the side surface of the second level interlayer insulating layer **23B** can be conductorized, thereby forming the second conductorization region **33C**. The second conductorization region **33C** is a region that is conductorized by the diffusion of hydrogen contained in the second level interlayer insulating layer **23B** into the oxide semiconductor, and can contact the side surface of the second level interlayer insulating layer **23B**.

[0159] In another portion of the second active pattern **33** in contact with the side surfaces of low hydrogen films, such as the second level active buffer layer **21B**, the first gate insulating layer **22**, and the first level interlayer insulating layer **23A**, the oxide semiconductor is not conductorized. Accordingly, the first oxide semiconductor region **33D** can be in contact with the side surfaces of the second level active buffer layer **21B**, the first gate insulating layer **22**, and the first level interlayer insulating layer **23A**.

[0160] The second active pattern **33** can further include an extension region **33B**. The extension region **33B** can be disposed on the second electrode **32** and can be in direct contact with the second

electrode **32**. The extension region **33B** can have different characteristics depending on the type of metallic material of the second electrode **32**. The extension region **33B** can be an oxide semiconductor region or a conductorization region in which an oxide semiconductor is conductorized.

[0161] In an example, the second electrode **32** can be formed of a metal having a higher oxidizing power than the oxide semiconductor. For example, the second electrode **32** can be formed of at least one of Ti, Al, MoTi, ITO, IZO, or Zr. In such a case, the metal of the second electrode **32** can be combined with the oxygen contained in the oxide semiconductor at the contact portion between the second electrode **32** and the oxide semiconductor, so that the oxide semiconductor can lose oxygen and be conductorized. In such a case, the extension region **33B** can be a conductorization region. In another example, the second electrode **32** can be formed of a metal having a lower oxidizing power than the oxide semiconductor. For example, the second electrode **32** can be formed of at least one of Mo, Cu, W, or Fe. In such a case, the metal of the second electrode **32** may not be combined with the oxygen contained in the oxide semiconductor at the contact portion between the second electrode **32** and the oxide semiconductor, so that the oxide semiconductor can remain in the oxide semiconductor state without being conductorized. In such a case, the extension region **33B** can be an oxide semiconductor region.

[0162] Referring to FIG. 7, each of the active buffer layer **21**, the first gate insulating layer **22**, and the interlayer insulating layer **23** can be formed of an insulating material, but the active buffer layer **21** and the interlayer insulating layer **23** can be formed of a material having a different hydrogen content than the first gate insulating layer **22**. The active buffer layer **21** and the interlayer insulating layer **23** can be formed of a high hydrogen film material having a hydrogen content equal to or higher than the threshold value, and the first gate insulating layer **22** can be formed of a low hydrogen film material having a hydrogen content lower than the threshold value. The active buffer layer **21** and the interlayer insulating layer **23** can have a higher hydrogen content than the first gate insulating layer **22**.

[0163] For example, the active buffer layer **21** and the interlayer insulating layer **23** can be formed of at least one selected from the group of materials including silicon nitride (SiN_x), aluminum oxide (AlO_x), hafnium oxide (HfO_x), zirconium oxide (ZrO_x), tantalum oxide (TaO_x), or the like, and the first gate insulating layer **22** can be formed of silicon oxide (SiO_x).

[0164] Here, the silicon nitride can be formed using silane (SiH_4) and ammonia (NH_3) as source gases. Aluminum oxide (AlO_x) can be formed using trimethyl aluminum (TMA, $\text{C}_3\text{H}_9\text{Al}$) as a source gas. Hafnium oxide (HfO_x) can be formed using CpHf ($\text{C}_{11}\text{H}_{23}\text{N}_3\text{Hf}$) as a source gas. Zirconium oxide (ZrO_x) can be formed using CpZr ($\text{C}_{11}\text{H}_{23}\text{N}_3\text{Hf}$) as a source gas. Tantalum oxide (TaO_x) can be formed using CpTa ($\text{C}_{17}\text{H}_{39}\text{N}_4\text{Ta}$) as a source gas.

[0165] In response to the diffusion of hydrogen from the active buffer layer **21**, which is a high hydrogen film, the oxide semiconductor in a portion of the second active pattern **33** in contact with the side surface of the active buffer layer **21** can be conductorized, thereby forming the first conductorization region **33A**. The first conductorization region **33A** is a region that is conductorized by the diffusion of hydrogen contained in the active buffer layer **21** into the oxide semiconductor, and can contact the side surface of the active buffer layer **21**.

[0166] In response to the diffusion of hydrogen from the interlayer insulating layer **23**, which is a high hydrogen film, the oxide semiconductor in another portion of the second active pattern **33** in contact with the side surface of the interlayer insulating layer **23** can be conductorized, thereby forming the second conductorization region **33C**. The second conductorization region **33C** is a region that is conductorized by the diffusion of hydrogen contained in the interlayer insulating layer **23** into the oxide semiconductor, and can contact the side surface of the interlayer insulating layer **23**.

[0167] In another portion of the second active pattern **33** in contact with the side surface of the first

gate insulating layer **22**, which is a low hydrogen film, the oxide semiconductor is not conductorized. Accordingly, the first oxide semiconductor region **33D** can be in contact with the side surface of the first gate insulating layer **22**.

[0168] In the embodiment illustrated with reference to FIG. **6**, the active buffer layer **21** has a structure in which a high hydrogen film and a low hydrogen film are stacked sequentially and the interlayer insulating layer **23** has a structure in which a low hydrogen film and a high hydrogen film are stacked sequentially, and in the embodiment illustrated with reference to FIG. **7**, the active buffer layer **21** and the interlayer insulating layer **23** are implemented as a high hydrogen film, but this is not intended to be limiting.

[0169] The active buffer layer **21** can have a structure in which a high hydrogen film and a low hydrogen film are stacked sequentially and the interlayer insulating layer **23** can have a single-layer structure of a high hydrogen film, or the active buffer layer **21** can have a single-layer structure of a high hydrogen film and the interlayer insulating layer **23** can have a structure in which a low hydrogen film and a high hydrogen film are stacked sequentially. In other words, the active buffer layer **21** can have a structure in which a high hydrogen film and a low hydrogen film are stacked sequentially or a single-layer structure of a high hydrogen film, and the interlayer insulating layer **23** can have a structure in which a low hydrogen film and a high hydrogen film are stacked sequentially or a single-layer structure of a high hydrogen film.

[0170] FIGS. **8A** to **8F** are cross-sectional views illustrating a method of manufacturing an oxide semiconductor TFT array substrate included in a stretchable display device according to embodiments of the present disclosure, and illustrate a method of manufacturing the oxide semiconductor TFT array substrate shown in FIG. **5**.

[0171] Referring to FIG. **8A**, a sacrificial layer **12**, a first substrate **10A**, and a multi-buffer layer **11** can be formed sequentially over a support substrate **13**, and a first electrode **31** of a second TFT can be formed over the multi-buffer layer **11** in a pixel area PIXA. In addition or optionally, a bottom shield metal pattern **91** can be further formed over the multi-buffer layer **11** in the pixel area PIXA. The support substrate **13** can serve to support the first substrate **10A** during the manufacturing process. The sacrificial layer **12** can bond the support substrate **13** and the first substrate **10A** together. The sacrificial layer **12** can be formed of an adhesive material, the adhesion of which can be reduced by at least one of a chemical treatment or an optical treatment. After completion of the manufacturing process, the adhesion of the sacrificial layer **12** can be reduced, and the sacrificial layer **12** and the support substrate **13** can be removed.

[0172] The first electrode **31** and the bottom shield metal pattern **91** can be formed together. The first electrode **31** and the bottom shield metal pattern **91** can be formed of the same material.

[0173] An active buffer layer **21** covering the first electrode **31** and the bottom shield metal pattern **91** can be formed over the multi-buffer layer **11**. In an embodiment, the active buffer layer **21** can be implemented as a low hydrogen film having a hydrogen content lower than a threshold value. In another embodiment, the active buffer layer **21** can have a structure in which a high hydrogen film having a hydrogen content equal to or higher than the threshold value and a low hydrogen film having a hydrogen content lower than a threshold are stacked sequentially. In another embodiment, the active buffer layer **21** can have a single-layer structure of a high hydrogen film having a hydrogen content equal to or higher than the threshold value. The high hydrogen film can be formed of at least one selected from the group of materials including silicon nitride (SiNx), aluminum oxide (AlOx), hafnium oxide (HfOx), zirconium oxide (ZrOx), tantalum oxide (TaOx), or the like. The hydrogen film can be formed of silicon oxide (SiOx).

[0174] The first active pattern **81** of the first TFT **80** can be formed over the active buffer layer **21** in the pixel area PIXA. In an example, the first active pattern **81** can be formed of an oxide semiconductor. In another example, the first active pattern **81** can be formed of amorphous silicon (a-Si), polycrystalline silicon (poly-Si), an organic semiconductor, or the like.

[0175] A first gate insulating layer **22** covering the first active pattern **81** can be formed over the

active buffer layer **21**. The first gate insulating layer **22** can be implemented as a low hydrogen film. The first gate insulating layer **22** can include silicon oxide (SiOx).

[0176] A first gate electrode **82** can be formed over the first gate insulating layer **22** in the pixel area PIXA to overlap a channel region of the first active pattern **81**. In addition or optionally, a capacitor bottom electrode **41** can be formed over the first gate insulating layer **22** in the pixel area PIXA. In addition or optionally, a first pad layer **51** can be formed over the first gate insulating layer **22** in a pad area PADA.

[0177] The first gate electrode **82**, the capacitor bottom electrode **41**, and the first pad layer **51** can be formed at one time. The first gate electrode **82**, the capacitor bottom electrode **41**, and the first pad layer **51** can be formed of the same material.

[0178] An interlayer insulating layer **23** can be formed over the first gate insulating layer **22** to cover the first gate electrode **82**, the capacitor bottom electrode **41**, and the first pad layer **51**.

[0179] In an embodiment, the interlayer insulating layer **23** can be implemented as a low hydrogen film having a hydrogen content lower than the threshold value. In another embodiment, the interlayer insulating layer **23** can have a structure in which a low hydrogen film having a hydrogen content lower than the threshold value and a high hydrogen film having a hydrogen content higher than the threshold value are stacked sequentially. In another embodiment, the interlayer insulating layer **23** can be implemented as a single layer of a high hydrogen film having a hydrogen content higher than the threshold value. The high hydrogen film can be formed of at least one selected from the group of materials including silicon nitride (SiNx), aluminum oxide (AlOx), hafnium oxide (HfOx), zirconium oxide (ZrOx), tantalum oxide (TaOx), or the like. The hydrogen film can include silicon oxide (SiOx).

[0180] The active buffer layer **21**, the first gate insulating layer **22**, and the interlayer insulating layer **23** can form the insulating member **20**.

[0181] Referring to FIG. **8B**, a first open area OP11 that penetrates a side surface of the interlayer insulating layer **23**, a side surface of the first gate insulating layer **22**, and the active buffer layer **21** can be formed to expose the stretchable area FA of the multi-buffer layer **11**. The first open area OP11 can be formed to expose a portion of the first electrode **31**. Through the first open area OP11, the side surface of the interlayer insulating layer **23**, a side surface of the first gate insulating layer **22**, and a side surface of the active buffer layer **21** can be exposed.

[0182] The interlayer insulating layer **23** can be formed to extend through a fourth open area OP21 to expose the first pad layer **51**. Through the fourth open area OP21, the top surface of the first pad layer **51** and the side surface of the interlayer insulating layer **23** can be exposed.

[0183] In addition, a first preliminary contact hole CH1 and a second preliminary contact hole CH2 that penetrate the interlayer insulating layer **23** and the first gate insulating layer **22** can be formed to expose a source region and a drain region of the first active pattern **81**.

[0184] Referring to FIG. **8C**, a capacitor top electrode **42**, a second electrode **32**, and a second pad layer **52** can be formed over the interlayer insulating layer **23**.

[0185] The capacitor top electrode **42** can be provided to overlap the capacitor bottom electrode **41**. The second electrode **32** can be disposed in the periphery of the first open area OP11. The second pad layer **52** can be formed over the top surface of the first pad layer **51** and the side surface of the interlayer insulating layer **23** exposed by the fourth open area OP21.

[0186] In an example, the second electrode **32** can be formed of a metal having a higher oxidizing power than the oxide semiconductor. For example, the second electrode **32** can include at least one of Ti, Al, MoTi, ITO, IZO, or Zr. In another example, the second electrode **32** can be formed of a metal having a lower oxidizing power than the oxide semiconductor. For example, the second electrode **32** can include at least one of Mo, Cu, W, or Fe. The capacitor top electrode **42** and the second pad layer **52** can be formed of the same material as the second electrode **32**.

[0187] Referring to FIG. **8D**, after a second active pattern **33** and a second gate insulating layer **34** are formed, a second open area OP12, a fifth open area OP22, a first contact hole CH1', and a

second contact hole CH2' can be formed.

[0188] The second active pattern **33** can be disposed over the second electrode **32** and extend to the first electrode **31** along the side surface of the interlayer insulating layer **23**, the side surface of the first gate insulating layer **22**, and the side surface of the active buffer layer **21** exposed by the first open area OP11. The top portion of the second active pattern **33** can be disposed over the second electrode **32** to directly contact the second electrode **32**. The bottom portion of the second active pattern **33** can be disposed over the first electrode **31** to directly contact the first electrode **31**. In an embodiment, the second active pattern **33** can be formed of an oxide semiconductor. In another embodiment, the second active pattern **33** can include a conductorization region.

[0189] A second gate insulating layer **34** can be disposed over the interlayer insulating layer **23** to cover the second active pattern **33**.

[0190] The second open area OP12 that penetrates the second gate insulating layer **34** and the multi-buffer layer **11** can be formed to expose the stretchable area FA of the first substrate **10A**. The fifth open area OP22 that penetrates the second gate insulating layer **34** can be formed to expose the second pad layer **52**. The top surface of the second pad layer **52** and the side surface of the second gate insulating layer **34** can be exposed by the fifth open area OP22.

[0191] The first contact hole CH1' that penetrates the second gate insulating layer **34** can be formed to be connected to the first spare contact hole CH1. The second contact hole CH2' that penetrates the second gate insulating layer **34** can be formed to be connected to the second spare contact hole CH2.

[0192] Referring to FIG. **8E**, a source electrode **83**, a drain electrode **84**, a line **85**, a second gate electrode **35**, and a third pad layer **53** can be formed over the second gate insulating layer **34**.

[0193] The source electrode **83** can be connected to the source region of the first active pattern **81** through the first contact hole CH1'. The drain electrode **84** can be connected to the drain region of the first active pattern **81** through the second contact hole CH2'.

[0194] The second gate electrode **35** can be formed to overlap the second active pattern **33**. The second gate electrode **35** can be formed such that a portion thereof overlaps the side surface of the insulating member **20** exposed by the first open area OP11 and a portion of the second active pattern **33** disposed over the first electrode **31** and another portion thereof overlaps another portion of the second active pattern **33** disposed over the second electrode **32**.

[0195] The line **85** can be disposed in the pixel area PIXA, and can be electrically connected to a component disposed in the pixel area PIXA. The third pad layer **53** can be formed to cover the top surface of the second pad layer **52** and the side surface of the second gate insulating layer **34** exposed by the fifth open area OP22.

[0196] The source electrode **83**, the drain electrode **84**, the line **85**, the second gate electrode **35**, and the third pad layer **53** can be formed at one time. The source electrode **83**, the drain electrode **84**, the line **85**, the second gate electrode **35**, and the third pad layer **53** can be formed of the same material.

[0197] Referring to FIG. **8F**, a passivation layer **60** covering the source electrode **83**, the drain electrode **84**, the second gate electrode **35**, the line **85**, and the third pad layer **53** can be formed over the second gate insulating layer **34**.

[0198] Thereafter, a third open area OP13 penetrating the passivation layer **60** to expose the stretchable area FA of the first substrate **10A** and a sixth open area OP23 exposing the third pad layer **53** can be formed. In addition, a contact hole exposing the line **85** can be formed.

[0199] Thereafter, a signal line **70** can be formed. The signal line **70** can be connected to a line **85** through a contact hole exposing the line **85** and extend to the stretchable area FA of the first substrate **10A** along a side surface of the passivation layer **60** exposed by the third open area OP13. In addition, the signal line **70** can extend to a pad area PADA and be connected to the third pad layer **53** through the sixth open area OP23.

[0200] A case in which the oxide semiconductor TFT array substrate is used in a stretchable display

device has been described hereinabove, this is not intended to be limiting. The oxide semiconductor TFT array substrate according to embodiments of the present disclosure can also be used in a non-stretchable display device. In such cases, the areas exposed by the first, second, and third open areas OP11, OP12, and OP13 in FIGS. 3, 5, and 8A through 8F may not be stretchable areas. [0201] The oxide semiconductor TFT array substrate and the stretchable display device including the same according to embodiments of the present disclosure described above can be briefly reviewed as follows.

[0202] A stretchable display device according to embodiments of the present disclosure includes a substrate including a pixel area and a stretchable area; an insulating member disposed over the substrate in the pixel area and including a first open area exposing the stretchable area; and a TFT disposed over an area including a side surface of the insulating member exposed by the first open area.

[0203] According to embodiments of the present disclosure, the TFT can include a first electrode disposed over the substrate in the pixel area, with at least a portion thereof being exposed by the first open area; a second electrode disposed over the insulating member above the first electrode to be adjacent to the first open area; an active pattern disposed over the second electrode and extending to the first electrode along the side surface of the insulating member exposed by the first open area; a gate insulating layer disposed over the insulating member, covering the active pattern, and including a second open area exposing the stretchable area; and a gate electrode disposed over the gate insulating layer to overlap the active pattern.

[0204] According to embodiments of the present disclosure, the active pattern can include an oxide semiconductor, and includes a first conductorization region in contact with the first electrode; a second conductorization region adjacent to the second electrode; and a first oxide semiconductor region between the first conductorization region and the second conductorization region.

[0205] According to embodiments of the present disclosure, the active pattern can further include a third conductorization region in contact with the second electrode and the second conductorization region.

[0206] According to embodiments of the present disclosure, the second electrode can include a metal having a higher oxidizing power than the oxide semiconductor.

[0207] According to embodiments of the present disclosure, the second electrode can include at least one of Ti, Al, MoTi, ITO, IZO, or Zr.

[0208] According to embodiments of the present disclosure, the active pattern can further include a second oxide semiconductor region in contact with the second electrode and the second conductorization region.

[0209] According to embodiments of the present disclosure, the second electrode can include a metal having a lower oxidizing power than the oxide semiconductor.

[0210] According to embodiments of the present disclosure, the second electrode can include at least one of Mo, Cu, W, or Fe.

[0211] According to embodiments of the present disclosure, the insulating member can include a first insulating layer disposed over the substrate; a second insulating layer disposed over the first insulating layer and having a lower hydrogen content than the first insulating layer; and a third insulating layer disposed over the second insulating layer and having a higher hydrogen content than the second insulating layer, wherein a side surface of the first insulating layer is in contact with the first conductorization region of the active pattern, a side surface of the second insulating layer is in contact with the first oxide semiconductor region of the active pattern, and a side surface of the third insulating layer is in contact with the second conductorization region of the active pattern.

[0212] According to embodiments of the present disclosure, each of the first insulating layer and the third insulating layer can include at least one selected from a group including silicon nitride (SiNx), aluminum oxide (AlOx), hafnium oxide (HfOx), zirconium oxide (ZrOx), or tantalum

oxide (TaOx), and the second insulating layer can include silicon oxide (SiOx).

[0213] According to embodiments of the present disclosure, the stretchable display device can further include a passivation layer disposed over the gate insulating layer, covering the gate electrode, and including a third open area exposing the stretchable area; and a signal line disposed over the passivation layer and extending to the stretchable area along a side surface of the passivation layer exposed by the third open area.

[0214] A stretchable display device according to embodiments of the present disclosure includes a substrate including a pixel area, a stretchable area, and a pad area; a buffer layer disposed over the substrate; a first active pattern disposed over the pixel area of the buffer layer, a first gate insulating layer disposed over the buffer layer and covering the first active pattern; a first gate electrode disposed over the first gate insulating layer to overlap a portion of the first active pattern; an interlayer insulating layer disposed over the first gate insulating layer and covering the first gate electrode; a first open area penetrating the interlayer insulating layer, the first gate insulating layer, and the buffer layer to expose the stretchable area of the substrate; a first electrode disposed over the pixel area of the substrate, with at least a portion thereof being exposed by the first open area; a second electrode disposed over the interlayer insulating layer above the first electrode to be adjacent to the first open area; a second active pattern disposed over the second electrode and extending to the first electrode along a side surface of the interlayer insulating layer, a side surface of the first gate insulating layer, and a side surface of the buffer layer exposed by the first open area; a second gate insulating layer disposed over the interlayer insulating layer, covering the second active pattern, and including a second open area exposing the stretchable area of the substrate; and a second gate electrode disposed over the second gate insulating layer to overlap the second active pattern.

[0215] According to embodiments of the present disclosure, the second active pattern can include an oxide semiconductor, and can include a first conductorization region in contact with the first electrode; a second conductorization region adjacent to the second electrode; and a first oxide semiconductor region between the first conductorization region and the second conductorization region.

[0216] According to embodiments of the present disclosure, the second active pattern can further include a third conductorization region in contact with the second electrode and the second conductorization region.

[0217] According to embodiments of the present disclosure, the second electrode can include a metal having a higher oxidizing power than the oxide semiconductor.

[0218] According to embodiments of the present disclosure, the second electrode can include at least one of Ti, Al, MoTi, ITO, IZO, or Zr.

[0219] According to embodiments of the present disclosure, the second active pattern can further include a second oxide semiconductor region in contact with the second electrode and the second conductorization region.

[0220] According to embodiments of the present disclosure, the second electrode can include a metal having a lower oxidizing power than the oxide semiconductor.

[0221] According to embodiments of the present disclosure, the second electrode can include at least one of Mo, Cu, W, or Fe.

[0222] According to embodiments of the present disclosure, the buffer layer can include a first level buffer layer disposed over the first electrode and the substrate and having a higher hydrogen content than the first gate insulating layer; and a second level buffer layer disposed between the first level buffer layer and the first gate insulating layer and having a lower hydrogen content than the first level buffer layer. A side surface of the first level buffer layer can be in contact with the first conductorization region of the second active pattern, and a side surface of the second level buffer layer and a side surface of the first gate insulating layer can be in contact with the first oxide semiconductor region of the second active pattern.

[0223] According to embodiments of the present disclosure, the interlayer insulating layer can include a first level interlayer insulating layer disposed over the first gate insulating layer; and a second level interlayer insulating layer disposed between the first level interlayer insulating layer and the second gate insulating layer, and having a higher hydrogen content than the first gate insulating layer and the first level interlayer insulating layer. The side surface of the first gate insulating layer and a side surface of the first level interlayer insulating layer can be in contact with the first oxide semiconductor region of the second active pattern. A side surface of the second level interlayer insulating layer can be in contact with the second conductorization region of the second active pattern.

[0224] According to embodiments of the present disclosure, the hydrogen content of the buffer layer can be higher than the hydrogen content of the first gate insulating layer, the side surface of the buffer layer can be in contact with the first conductorization region of the second active pattern, and the side surface of the first gate insulating layer can be in contact with the first oxide semiconductor region of the second active pattern.

[0225] According to embodiments of the present disclosure, the hydrogen content of the interlayer insulating layer can be higher than the hydrogen content of the first gate insulating layer, the side surface of the interlayer insulating layer can be in contact with the second conductorization region of the second active pattern, and the side surface of the first gate insulating layer can be in contact with the first oxide semiconductor region of the second active pattern.

[0226] According to embodiments of the present disclosure, the stretchable display device can further include a bottom shield metal pattern disposed over the substrate and overlapping the first active pattern. The first electrode can include the same material as the bottom shield metal pattern.

[0227] According to embodiments of the present disclosure, the stretchable display device can further include a capacitor bottom electrode disposed over the first gate insulating layer; and a capacitor top electrode disposed over the interlayer insulating layer and overlapping the capacitor bottom electrode. The second electrode can include the same material as the capacitor top electrode.

[0228] According to embodiments of the present disclosure, the stretchable display device can further include a capacitor bottom electrode disposed over the first gate insulating layer; and a capacitor top electrode disposed over the interlayer insulating layer and overlapping the capacitor bottom electrode. The first gate electrode includes the same material as the capacitor bottom electrode.

[0229] According to embodiments of the present disclosure, the stretchable display device can further include a source electrode disposed over the second gate insulating layer and connected to a source region of the first active pattern through a first contact hole penetrating the second gate insulating layer, the interlayer insulating layer, and the first gate insulating layer; and a drain electrode disposed over the second gate insulating layer and connected to a drain region of the first active pattern through a second contact hole penetrating the second gate insulating layer, the interlayer insulating layer, and the first gate insulating layer. The second gate electrode can include the same material as the source electrode and the drain electrode.

[0230] According to embodiments of the present disclosure, the stretchable display device can further include a source electrode disposed over the second gate insulating layer and connected to a source region of the first active pattern through a first contact hole penetrating the second gate insulating layer, the interlayer insulating layer, and the first gate insulating layer, and a drain electrode disposed over the second gate insulating layer and connected to a drain region of the first active pattern through a second contact hole penetrating the second gate insulating layer, the interlayer insulating layer, and the first gate insulating layer. A portion of the second gate electrode can be disposed on the same layer as the source electrode and the drain electrode.

[0231] According to embodiments of the present disclosure, the stretchable display device can further include a pad disposed in the pad area. The pad can include a first pad layer disposed over

the first gate insulating layer, and a second pad layer disposed over the first pad layer exposed by a third open area penetrating the interlayer insulating layer.

[0232] According to embodiments of the present disclosure, the second electrode can include the same material as the second pad layer.

[0233] According to embodiments of the present disclosure, the pad can further include a third pad layer disposed over the second pad layer exposed by a fourth open area penetrating the second gate insulating layer. The second gate electrode can include the same material as the third pad layer.

[0234] A TFT array substrate according to embodiments of the present disclosure includes a buffer layer disposed over a substrate; a first gate insulating layer disposed over the buffer layer, an interlayer insulating layer disposed over the first gate insulating layer; a first open area penetrating the interlayer insulating layer, the first gate insulating layer, and the buffer layer to expose a portion of the substrate; a first TFT; and a second TFT. The first TFT can include a first oxide semiconductor pattern disposed over the buffer layer and covered with the first gate insulating layer; the first gate insulating layer, and a first gate electrode disposed over the first gate insulating layer to overlap a portion of the first oxide semiconductor pattern. The second TFT can include a first electrode disposed over the substrate, with at least a portion thereof being exposed by the first open area; a second electrode disposed over the interlayer insulating layer above the first electrode to be adjacent to the first open area; a second oxide semiconductor pattern disposed over the second electrode, and extending to the first electrode along a side surface of the interlayer insulating layer, a side surface of the first gate insulating layer, and a side surface of the buffer layer exposed by the first open area; a second gate insulating layer disposed over the interlayer insulating layer and covering the second oxide semiconductor pattern; and a second gate electrode disposed over the second gate insulating layer to overlap the second oxide semiconductor pattern.

[0235] According to embodiments of the present disclosure having the above-described structure, in the oxide semiconductor TFT array substrate and the stretchable display device including the same, a TFT can be disposed on a side surface of the insulating member exposed by the first open area exposing the stretchable region to reduce the area occupied by the TFT, thereby reducing the size of the pixel area and improving resolution.

[0236] According to embodiments of the present disclosure, in the oxide semiconductor TFT array substrate and the stretchable display device including the same, the size of the pixel area can be reduced by reducing the area occupied by the TFT, and the free area resulting from the reduced pixel area can be used as a stretchable area to improve stretching characteristics.

[0237] According to embodiments of the present disclosure, in the oxide semiconductor TFT array substrate and the stretchable display device including the same, the impact of stretching stress on the TFT can be reduced to reduce damage to the TFT and reduce or prevent characteristics deterioration of the TFT.

[0238] According to embodiments of the present disclosure, in the oxide semiconductor TFT array substrate and the stretchable display device including the same, the oxide semiconductor can be conductorized by diffusing ions contained in the insulating member to form a conductorization region, thereby reducing the number of process steps. Accordingly, the effect of process optimization can be obtained.

[0239] The above description has been presented to enable any person skilled in the art to make and use the technical idea of the present disclosure, and has been provided in the context of a particular application and its requirements. Various modifications, additions and substitutions to the described embodiments of the present disclosure will be readily apparent to those skilled in the art, and the general principles defined herein can be applied to other embodiments of the present disclosure and applications without departing from the spirit and scope of the present disclosure. The above description and the accompanying drawings provide an example of the technical idea of the present disclosure for illustrative purposes only. That is, the disclosed embodiments are intended to illustrate the scope of the technical idea of the present disclosure.

Claims

1. A stretchable display device comprising: a substrate comprising a pixel area and a stretchable area; an insulating member disposed over the substrate in the pixel area and comprising a first open area exposing the stretchable area; and a thin-film transistor disposed over an area including a side surface of the insulating member exposed by the first open area.
2. The stretchable display device of claim 1, wherein the thin-film transistor comprises: a first electrode disposed over the substrate in the pixel area, with at least a portion of the first electrode being exposed by the first open area; a second electrode disposed over the insulating member above the first electrode to be adjacent to the first open area; an active pattern disposed over the second electrode and extending to the first electrode along the side surface of the insulating member exposed by the first open area; a gate insulating layer disposed over the insulating member, covering the active pattern, and comprising a second open area exposing the stretchable area; and a gate electrode disposed over the gate insulating layer to overlap the active pattern.
3. The stretchable display device of claim 2, wherein the active pattern comprises an oxide semiconductor, and comprises: a first conductorization region in contact with the first electrode; a second conductorization region adjacent to the second electrode; and a first oxide semiconductor region between the first conductorization region and the second conductorization region.
4. The stretchable display device of claim 3, wherein the active pattern further comprises a third conductorization region in contact with the second electrode and the second conductorization region.
5. The stretchable display device of claim 4, wherein the second electrode comprises a metal having a higher oxidizing power than the oxide semiconductor.
6. The stretchable display device of claim 4, wherein the second electrode comprises at least one of Ti, Al, MoTi, ITO, IZO, or Zr.
7. The stretchable display device of claim 3, wherein the active pattern further comprises a second oxide semiconductor region in contact with the second electrode and the second conductorization region.
8. The stretchable display device of claim 7, wherein the second electrode comprises a metal having a lower oxidizing power than the oxide semiconductor.
9. The stretchable display device of claim 7, wherein the second electrode comprises at least one of Mo, Cu, W, or Fe.
10. The stretchable display device of claim 3, wherein the insulating member comprises: a first insulating layer disposed over the substrate; a second insulating layer disposed over the first insulating layer and having a lower hydrogen content than the first insulating layer, and a third insulating layer disposed over the second insulating layer and having a higher hydrogen content than the second insulating layer, wherein a side surface of the first insulating layer is in contact with the first conductorization region of the active pattern, a side surface of the second insulating layer is in contact with the first oxide semiconductor region of the active pattern, and a side surface of the third insulating layer is in contact with the second conductorization region of the active pattern.
11. The stretchable display device of claim 10, wherein each of the first insulating layer and the third insulating layer comprises at least one selected from a group including silicon nitride (SiNx), aluminum oxide (AlOx), hafnium oxide (HfOx), zirconium oxide (ZrOx), or tantalum oxide (TaOx), and wherein the second insulating layer comprises silicon oxide (SiOx).
12. The stretchable display device of claim 2, further comprising: a passivation layer disposed over the gate insulating layer, covering the gate electrode, and comprising a third open area exposing the stretchable area; and a signal line disposed over the passivation layer and extending to the stretchable area along a side surface of the passivation layer exposed by the third open area.

13. A stretchable display device comprising: a substrate comprising a pixel area, a stretchable area, and a pad area; a buffer layer disposed over the substrate; a first active pattern disposed over the pixel area of the buffer layer; a first gate insulating layer disposed over the buffer layer and covering the first active pattern; a first gate electrode disposed over the first gate insulating layer to overlap a portion of the first active pattern; an interlayer insulating layer disposed over the first gate insulating layer and covering the first gate electrode; a first open area penetrating the interlayer insulating layer, the first gate insulating layer, and the buffer layer to expose the stretchable area of the substrate; a first electrode disposed over the pixel area of the substrate, with at least a portion of the first electrode being exposed by the first open area; a second electrode disposed over the interlayer insulating layer above the first electrode to be adjacent to the first open area; a second active pattern disposed over the second electrode and extending to the first electrode along a side surface of the interlayer insulating layer, a side surface of the first gate insulating layer, and a side surface of the buffer layer exposed by the first open area; a second gate insulating layer disposed over the interlayer insulating layer, covering the second active pattern, and comprising a second open area exposing the stretchable area of the substrate; and a second gate electrode disposed over the second gate insulating layer to overlap the second active pattern.

14. The stretchable display device of claim 13, wherein the second active pattern comprises an oxide semiconductor, and comprises: a first conductorization region in contact with the first electrode; a second conductorization region adjacent to the second electrode; and a first oxide semiconductor region between the first conductorization region and the second conductorization region.

15. The stretchable display device of claim 14, wherein the second active pattern further comprises a third conductorization region in contact with the second electrode and the second conductorization region.

16. The stretchable display device of claim 15, wherein the second electrode comprises a metal having a higher oxidizing power than the oxide semiconductor.

17. The stretchable display device of claim 15, wherein the second electrode comprises at least one of Ti, Al, MoTi, ITO, IZO, or Zr.

18. The stretchable display device of claim 14, wherein the second active pattern further comprises a second oxide semiconductor region in contact with the second electrode and the second conductorization region.

19. The stretchable display device of claim 18, wherein the second electrode comprises a metal having a lower oxidizing power than the oxide semiconductor.

20. The stretchable display device of claim 18, wherein the second electrode comprises at least one of Mo, Cu, W, or Fe.

21. The stretchable display device of claim 14, wherein the buffer layer comprises: a first level buffer layer disposed over the first electrode and the substrate and having a higher hydrogen content than the first gate insulating layer; and a second level buffer layer disposed between the first level buffer layer and the first gate insulating layer and having a lower hydrogen content than the first level buffer layer, wherein a side surface of the first level buffer layer is in contact with the first conductorization region of the second active pattern, and wherein a side surface of the second level buffer layer and a side surface of the first gate insulating layer are in contact with the first oxide semiconductor region of the second active pattern.

22. The stretchable display device of claim 14, wherein the interlayer insulating layer comprises: a first level interlayer insulating layer disposed over the first gate insulating layer, and a second level interlayer insulating layer disposed between the first level interlayer insulating layer and the second gate insulating layer, and having a higher hydrogen content than the first gate insulating layer and the first level interlayer insulating layer, wherein the side surface of the first gate insulating layer and a side surface of the first level interlayer insulating layer are in contact with the first oxide semiconductor region of the second active pattern, and wherein a side surface of the second level

interlayer insulating layer is in contact with the second conductorization region of the second active pattern.

23. The stretchable display device of claim 14, wherein the hydrogen content of the buffer layer is higher than the hydrogen content of the first gate insulating layer, wherein the side surface of the buffer layer is in contact with the first conductorization region of the second active pattern, and wherein the side surface of the first gate insulating layer is in contact with the first oxide semiconductor region of the second active pattern.

24. The stretchable display device of claim 14, wherein the hydrogen content of the interlayer insulating layer is higher than the hydrogen content of the first gate insulating layer, wherein the side surface of the interlayer insulating layer is in contact with the second conductorization region of the second active pattern, and wherein the side surface of the first gate insulating layer is in contact with the first oxide semiconductor region of the second active pattern.

25. The stretchable display device of claim 13, further comprising a bottom shield metal pattern disposed over the substrate and overlapping the first active pattern, wherein the first electrode comprises a same material as the bottom shield metal pattern.

26. The stretchable display device of claim 13, further comprising: a capacitor bottom electrode disposed over the first gate insulating layer; and a capacitor top electrode disposed over the interlayer insulating layer and overlapping the capacitor bottom electrode, wherein the second electrode comprises a same material as the capacitor top electrode.

27. The stretchable display device of claim 13, further comprising: a capacitor bottom electrode disposed over the first gate insulating layer; and a capacitor top electrode disposed over the interlayer insulating layer and overlapping the capacitor bottom electrode, wherein the first gate electrode comprises a same material as the capacitor bottom electrode.

28. The stretchable display device of claim 13, further comprising: a source electrode disposed over the second gate insulating layer and connected to a source region of the first active pattern through a first contact hole penetrating the second gate insulating layer, the interlayer insulating layer, and the first gate insulating layer, and a drain electrode disposed over the second gate insulating layer and connected to a drain region of the first active pattern through a second contact hole penetrating the second gate insulating layer, the interlayer insulating layer, and the first gate insulating layer, wherein the second gate electrode comprises a same material as the source electrode and the drain electrode.

29. The stretchable display device of claim 13, further comprising: a source electrode disposed over the second gate insulating layer and connected to a source region of the first active pattern through a first contact hole penetrating the second gate insulating layer, the interlayer insulating layer, and the first gate insulating layer, and a drain electrode disposed over the second gate insulating layer and connected to a drain region of the first active pattern through a second contact hole penetrating the second gate insulating layer, the interlayer insulating layer, and the first gate insulating layer, wherein a portion of the second gate electrode is disposed on a same layer as the source electrode and the drain electrode.

30. The stretchable display device of claim 13, further comprising a pad disposed in the pad area, wherein the pad comprises: a first pad layer disposed over the first gate insulating layer; and a second pad layer disposed over the first pad layer exposed by a third open area penetrating the interlayer insulating layer.

31. The stretchable display device of claim 30, wherein the second electrode comprises a same material as the second pad layer.

32. The stretchable display device of claim 30, wherein the pad further comprises a third pad layer disposed over the second pad layer exposed by a fourth open area penetrating the second gate insulating layer, and wherein the second gate electrode comprises a same material as the third pad layer.

33. A thin-film transistor array substrate comprising: a buffer layer disposed over a substrate; a first

gate insulating layer disposed over the buffer layer, an interlayer insulating layer disposed over the first gate insulating layer; a first open area penetrating the interlayer insulating layer, the first gate insulating layer, and the buffer layer to expose a portion of the substrate; a first thin-film transistor; and a second thin-film transistor, wherein the first thin-film transistor comprises: a first oxide semiconductor pattern disposed over the buffer layer and covered with the first gate insulating layer, the first gate insulating layer; and a first gate electrode disposed over the first gate insulating layer to overlap a portion of the first oxide semiconductor pattern, and wherein the second thin-film transistor comprises: a first electrode disposed over the substrate, with at least a portion of the first electrode being exposed by the first open area; a second electrode disposed over the interlayer insulating layer above the first electrode to be adjacent to the first open area; a second oxide semiconductor pattern disposed over the second electrode, and extending to the first electrode along a side surface of the interlayer insulating layer, a side surface of the first gate insulating layer, and a side surface of the buffer layer exposed by the first open area; a second gate insulating layer disposed over the interlayer insulating layer and covering the second oxide semiconductor pattern; and a second gate electrode disposed over the second gate insulating layer to overlap the second oxide semiconductor pattern.
